Supporting information for *Scheduling ^{225}Ac purification under isotope and delivery uncertainty using DWSIM process modeling*

S1. Feasibility and scheduling derivations

S1.1 Individual and combined impurity constraints

Let $a = \lambda_{226} - \lambda_{225} > 0$, $b = \lambda_{225} - \lambda_{227} > 0$, and let the strictly positive reference activity ratios be r_{226} and r_{227}. Common chemical transfer multiplies all actinium activities by the same factor, so

$$R_{226}(t) = r_{226}e^{-at} \tag{S1}$$

$$R_{227}(t) = r_{227}e^{bt} \tag{S2}$$

The first decreases and the second increases. Individual constraints $R_i \le u_i$ therefore imply the interval

$$L_{ind} = \max\{0, \ln(r_{226}/u_{226})/a\} \tag{S3}$$

$$U_{ind} = \ln(u_{227}/r_{227})/b \tag{S4}$$

The condition $L_{ind} \le U_{ind}$ is both necessary and sufficient. For $r_{226} > u_{226}$ it is equivalent to

$$r_{227} \le u_{227}(r_{226}/u_{226})^{-b/a} \tag{S5}$$

If $r_{226} \le u_{226}$, the condition reduces to $r_{227} \le u_{227}$ at the reference time.

For the actinium-only activity fraction P_{Ac}, define $q = P_{min}^{-1} - 1$. The combined condition is $f(t) \le q$, where

$$f(t) = r_{226}e^{-at} + r_{227}e^{bt} \tag{S6}$$

$$f''(t) = a^2 r_{226}e^{-at} + b^2 r_{227}e^{bt} > 0 \tag{S7}$$

Strict convexity gives at most two boundary times and a single interior minimum. The minimizer on $t \ge 0$ is

$$t_* = \max\{0, \ln(ar_{226}/br_{227})/(a+b)\} \tag{S8}$$

The code evaluates $f(t_*)$ before root finding. If this exceeds q, the interval is empty. If it equals q within the numerical tolerance, the interval is a single point. Otherwise Brent's method finds the lower root on $[0,t_*]$, unless $f(0) \le q$, and finds the upper root after doubling its bracket until $f > q$. The root tolerance is 10^{-12} d. The feasibility comparison tolerance is 10^{-13} in the dimensionless impurity sum. All reported conclusions lie well away from a numerically ambiguous feasibility classification except the deliberately constructed parent tangency.

S1.2 Robust window and objective

At any fixed time, f is increasing in each reference ratio. Thus, for rectangular uncertainty in the ratios, the simultaneous upper corner controls the impurity constraint. Write its feasible delivery interval as $[L^+,U^+]$. For start age c and delay $D \in [D_-,D_+]$, every possible delivery must lie inside this interval:

$$[c + D_-, c + D_+] \subseteq [L^+, U^+] \tag{S9}$$

This set inclusion proves the complete start window

$$\max(0, L^+ - D_-) \le c \le U^+ - D_+ \tag{S10}$$

Alternatively, convexity implies $f(t) \le \max\{f(c + D_-), f(c + D_+)\}$ over the delivery interval, so checking the two endpoint constraints suffices. Monotonicity in the initial ratios and convexity in delivery age establish a deterministic bound; a Monte Carlo pass fraction is not needed for the proof.

For fixed initial ^{225}Ac activity, the worst delivered activity is attained at the lowest admissible recovery and longest delay:

$$Y(c) = \eta_- e^{-\lambda_{225}(c+D_+)} \tag{S11}$$

Since $dY/dc < 0$, the optimum is the earliest feasible start $c_* = \max(0, L^+ - D_-)$. The general expression is

$$Y_* = \eta_- e^{-\lambda_{225}[\max(0,L^+-D_-)+D_+]}. \tag{S12}$$

For the studied case $L^+ > D_-$, it simplifies to

$$Y_* = \eta_- e^{-\lambda_{225}[L^+ + (D_+ - D_-)]}. \tag{S13}$$

If $L^+ \le D_-$, the start boundary at zero becomes active, and a uniform shift in both delay bounds can change the optimum delivered activity. The translation invariance discussed in the main article is restricted to the inactive-zero-boundary case.

With nominal earliest feasible delivery L_0 and nominal recovery η_0, the penalty is

$$\frac{Y_*}{\eta_0 e^{-\lambda_{225} L_0}} = \underbrace{\frac{\eta_-}{\eta_0}}_{\text{recovery}} \quad \underbrace{e^{-\lambda_{225}(L^+ - L_0)}}_{\text{isotope uncertainty}} \; \underbrace{e^{-\lambda_{225}(D_+ - D_-)}}_{\text{delay width}}. \tag{S14}$$

For the proton scenario the factors are 0.950000, 0.960317 and 0.869578. Their product is 0.793317. This comparison fixes the reference ^{225}Ac activity; uncertainty in its absolute assay is not included. The ratio rectangle is a design set, not a joint confidence region. For independent normal variables, a rectangle with two marginal standard deviations on each side would contain approximately $0.9545^2 = 0.9111$ of their joint distribution. This fact does not equal the scheduling pass fraction because feasibility need not fail immediately outside the rectangle.

S2. Parent retention and matrix formulation

The state in activity units is $A = (A_{225}, A_{226}, A_{227}, A_R)^T$. Before parent removal,

$$\frac{d\mathbf{A}}{dt} = M\mathbf{A}, \tag{S15}$$

$$M = \begin{pmatrix} -\lambda_{225} & 0 & 0 & \lambda_{225} \\ 0 & -\lambda_{226} & 0 & 0 \\ 0 & 0 & -\lambda_{227} & 0 \\ 0 & 0 & 0 & -\lambda_R \end{pmatrix}. \tag{S16}$$

The feeding coefficient in an activity matrix is λ_{225}, whereas the corresponding coefficient in an atom-population matrix is λ_R. Confusing these representations gives an incorrect parent contribution. An instantaneous separation at s is represented by $S = \mathrm{diag}(\eta, \eta, \eta, 0)$, so

$$\mathbf{A}_{\mathrm{del}} = e^{M(t-s)} S e^{Ms} \mathbf{A}_0. \tag{S17}$$

Its first component, normalized by $A_{225,0}$, agrees with

$$\eta e^{-\lambda_{225} t} \left[1 + \rho \frac{\lambda_{225}}{\lambda_{225} - \lambda_R} \left(e^{(\lambda_{225} - \lambda_R)s} - 1 \right) \right]. \tag{S18}$$

The implementation uses expm1 in CPython for numerical stability when the exponent is small. The source-consistent half-lives used here are distinct and do not require the equal-rate limit.

The critical parent calculation sets $s = t - 4$ d and searches $4 < t \leq 80$ d. It minimizes the required nonnegative parent activity ratio, with a bounded scalar minimizer and absolute timing tolerance 10^{-11} d. The nitrogen combined-target minimum is 0.560275 at $t = 17.594793$ d, well inside this interval. For individual caps the numerator is instead $\max\{R_{226}/0.01, R_{227}/0.01\} - 1$, giving 0.153239 at 12.881763 d. The supplied

time curves show the relevant minima and purity boundaries. These ratios refer to directly specified reference inventories; no parent activity is inferred from measured final recovery.

The evaluated ^{225}Ra decay dataset supplies the 14.9 d half-life used for the scenario (National Nuclear Data Center, 2022). The model does not follow ^{221}Fr, ^{217}At, ^{213}Bi or other daughters of the delivered actinium isotopes. A separate daughter-retention and dose model would be necessary for those questions.

S3. Numerical checks and prior published benchmark

S3.1 Verification protocol

The analysis uses CPython 3.11.9, NumPy 2.4.6 and SciPy 1.17.1. Two hundred randomized cases cover delivery ages 0.1–80 d, parent-removal ages between zero and delivery, r_{226} from 0.1 to 100, r_{227} from 10^{-4} to 0.1, parent ratios from zero to two and recoveries from 0.1 to one. Ratio ranges are sampled uniformly in logarithmic space. The matrix solution uses SciPy's matrix exponential. Independent numerical integration uses solve_ivp with relative tolerance 10^{-10} and absolute tolerance 10^{-13}, separately before and after the chemical event.

Table S1. Numerical verification results. These establish implementation consistency, not experimental process validity.

Check	Maximum error or result
Closed solution versus matrix exponential, relative	8.33×10^{-13}
Closed solution versus numerical ODE, scaled	3.82×10^{-7}
Tracked heavy-nucleus ledger closure, relative	1.16×10^{-15}
Matrix semigroup identity, absolute	9.96×10^{-15}
Cancellation of common recovery in ratios, absolute	7.11×10^{-15}
Native DWSIM versus CPython activity outputs, relative	1.60×10^{-16}
DWSIM solver errors; saved-file recalculation	0; verified
Minimum purity at ratio corners and 101 delivery delays	0.990000

The ODE error is $|A_{closed} - A_{ODE}|/\max(A_{closed}, 10^{-7})$ componentwise for unit initial ^{225}Ac activity; this avoids interpreting a tiny absolute error in almost fully decayed ^{226}Ac as a large relative error. The matrix relative-error denominator is $\max(A_{closed}, 10^{-14})$. The ledger tracks the four initial heavy nuclei and a lumped descendant population, one heavy descendant per tracked decay. It verifies the algebraic transfer of populations, not conservation of radioactive activity, total atomic mass, or a fully resolved decay chain including emitted particles.

The semigroup check compares $e^{M(t-s)}e^{Ms}$ and e^{Mt} without a separation event. The purity-ratio check verifies the cancellation of common chemical recovery in the parent-free model. The robust corner check evaluates all four reference-ratio corners at 101 evenly spaced delays from three to five days using the earliest

robust start. A separate calculation shifts both delay bounds earlier by 0.375 d and confirms the same optimized worst delivery age.

S3.2 Published processing-time example

Fitzsimmons et al. report a calculated proton-irradiated thorium example with actinium activities at EOB and seven days later (Fitzsimmons et al., 2019b). This prior work already examines purity as a function of processing time. We reproduce the example using its approximate half-lives: 10 d for ^{225}Ac, 29.37 h for ^{226}Ac and 21.772 y for ^{227}Ac. These values are used only for this benchmark; they do not replace the main-study inputs.

Table S2. Reproduction of the calculated activities in Fitzsimmons et al., Table 2. Values are in mCi; the benchmark is not an independent experimental observation.

Isotope	Source EOB	Source day 7	Model day 7	Difference (%)
^{225}Ac	206.91	127.37	127.36805	−0.0015
^{226}Ac	433.01	8.21	8.21428	+0.0521
^{227}Ac	0.42	0.42	0.41974	−0.0610

The rounded tabulated values account for the small differences. The model gives a maximum actinium-only purity of 99.3175% at 8.7324 d after the seven-day separation point, consistent with the source's approximately 99.3% and nine-day description. Figure S1 illustrates the reproduced calculation.

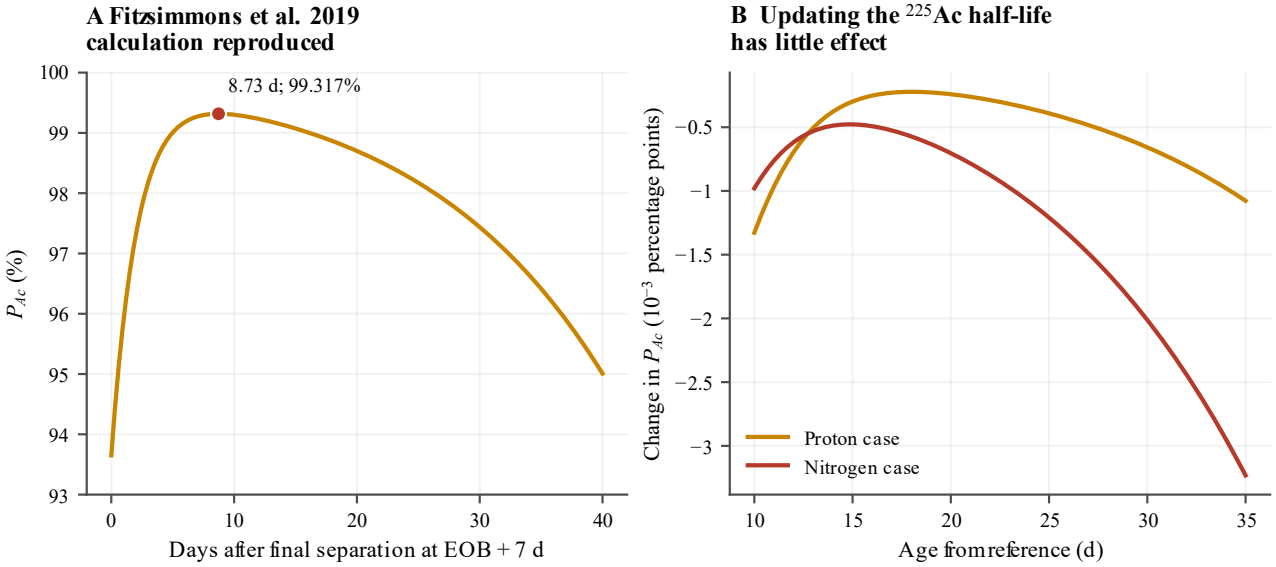


Figure S1. A: Reproduction of the established isotope-timing calculation using the reference inventory and half-lives of Fitzsimmons et al. The red marker is the maximum calculated here, measured from final separation rather than EOB. B: Change in the main-study purity curves when the ^{225}Ac half-life is changed from 9.92 to 9.9172 d, with all other inputs fixed. The latter sensitivity uses the two Tan et al. inventories and is independent of the prior benchmark.

S4. Source recovery and timing reconciliation

Table S3 records the recovery observations extracted from Tan et al., Table 2 (Tan et al., 2025). A mean of the reported percentage yields for runs 2-1 and 2-2 gives $\eta_{cal} = (0.639 + 0.643)/2 = 0.641$. This factor predicts product activity as $\hat{A}_{product} = \eta_{cal} A_{feed}$ on the source's reference-activity basis. It is checked against run 2-3 without fitting to that run.

Table S3. Recovery consistency calculation. Activities are in kBq.

Run	Feed	Product	Prediction	Residual (%)	Role
1-1b	5.8	2.7	3.7178	+37.70	Development
1-2	94.0	53.2	60.2540	+13.26	Development
2-1	46.3	29.6	29.6783	+0.26	Calibration
2-2	224.9	144.6	144.1609	−0.30	Calibration
2-3	164.4	102.9	105.3804	+2.41	Withheld

Residuals equal $(\hat{A} - A)/A$. The reported 46.7% yield attached to the early experiment is an average involving runs 1-1b and 1-1c, not simply the ratio of the displayed rounded 1-1b activities. That development result is excluded from calibration. The stabilized three-run average used in prospective scenarios is $(0.639 + 0.643 + 0.626)/3 = 0.636$, whereas the withheld-run check uses 0.641. This separation prevents claiming a holdout prediction after fitting to the holdout.

The source does not provide a complete common timestamp record for all reported activities. Consequently, interpreting 0.636 as a decay-corrected chemical transfer factor is an explicit prospective-model assumption. If an observed recovery already contained a nine-hour decay factor, treating it as pure chemical loss and then applying that same decay interval would underestimate a corrected prediction by the factor $e^{-\lambda_{225}(9/24)}$. The corrected prediction would be 2.65% higher than the double-counted one. This ambiguity affects absolute activity, while common transfer cancels from actinium isotope ratios.

The source's separate five-hour primary-separation and four-hour purification totals are used as stage durations. Its sixty-hour EOB-to-product statement for an example and the fourteen-day cooling entry for the corresponding target are not combined into an invented detailed schedule. For the baseline isotope analysis, the two Table 4 ratios are interpreted on the EOB-equivalent reference basis of the source's Fig. 11. Their parent-free evolution is not evidence that no radium was physically present in the target. The parent scenario therefore remains explicitly separate.

S5. Partial liquid recipe and counting rules

The methods describe a 10 mL AG50 cation-exchange bed, a 2 mL first DGA bed, a 5 mL second DGA bed and a 2 mL LN bed (Tan et al., 2025). Figure 7 of that source instead illustrates an aliquot using a 5 mL AG50 bed. The following ledger follows the methods-scale basis; it does not combine the aliquot bed

volume with the full recipe. Bed volumes are used only to translate reported liquid quantities. They do not calibrate resin capacity or retention kinetics.

Table S4. Documented fresh additions.

Stage	Operation	mL	HNO_3 (M)	HCl (M)	Source
Primary	DGA1 conditioning	10	5	0	p.2, Sec.2.3
Primary	AG50 rinse	50	0.05	0	p.2, Sec.2.3
Primary	Ac/Ra transfer	60	5	0	p.2, Sec.2.3
Primary	DGA1 Ac elution	80	10	0	pp.2,6
Primary	DGA1 lanthanide strip	16	0.05	0	p.2, Sec.2.3
Purification	DGA2 conditioning	25	5	0	p.3, Sec.2.4
Purification	Dilution water	80	0	0	p.3, Sec.2.4
Purification	DGA2 rinse	15	0	6	p.3, Sec.2.4
Purification	DGA2 Ac collection	10	0	0	p.3, Sec.2.4
Purification	LN conditioning	10	0	0.05	p.3, Sec.2.4
Purification	LN rinse	14	0.05	0	p.3, Sec.2.4
Purification	Final product elution	10	0.7	0	p.3, Sec.2.4

The sum of fresh additions is 380 mL. Analytical amounts are 1.286 mol HNO_3 and 0.0905 mol HCl; these are added formula-unit amounts, not equilibrium free-ion concentrations. The 80 mL primary actinium fraction subsequently passes through DGA2 and the 10 mL DGA2 actinium fraction subsequently passes through LN. These 90 mL are counted once more for pumping but are not counted again as fresh additions.

The resulting primary pumped subset is 216 mL, or 72.0 min at 3 mL min^{-1}. The purification subset is 164 mL of fresh additions plus 90 mL of transfers, or 84.67 min. Total documented pumping is 156.67 min. The difference from the reported nine-hour stage total is 383.33 min, or 70.99% of elapsed time. Missing target feed, sulfate operations, initial water, dead-volume rinses, pH adjustment and handling prevent interpreting this difference as idle time.

The software also reports 370 mL as the documented fresh volume outside the 10 mL final fraction. This is only an additive-volume bookkeeping quantity within the stated subset. It must not be called total radioactive waste: initial target liquid and other unquantified streams are absent, volume additivity is assumed, and physical hold-up and evaporation are not modelled. No environmental discharge or disposal calculation is made.

S6. Baseline native DWSIM workspace and reproducibility

The saved file dwsim/Ac225_Isotope_Timing.dwxml was created through DWSIM 10.2.4.0 Automation3. The six native custom operations are listed in Table S5. Each isotope node uses an arbitrary

reference ^{225}Ac activity of 10^6 Bq; all normalized quantities are independent of that scale. The native scripts are IronPython-compatible and are embedded in the workspace as well as supplied separately.

Table S5. Native custom-operation calculations.

Operation	Purpose
Proton case – parent free	Nominal isotope inventory without parent feeding
Proton case – ^{225}Ra scenario	Explicit reference parent-ratio scenario
Nitrogen case – parent free	Nominal isotope inventory without parent feeding
Nitrogen case – ^{225}Ra scenario	Explicit reference parent-ratio scenario
Proton robust upper input corner	Worst ratio corner for the stated uncertainty box
Documented liquid ledger	Fresh additions, internal transfer and acid inventory

These baseline operations are independent computational nodes. There are no connecting material streams and no thermodynamic property package in this baseline. The process sequence in main-text Fig. 1 is a conceptual boundary diagram, not a screenshot of a thermodynamically connected plant. DWSIM executes the inventories; CPython performs robust-window calculations and statistical analyses. This division is consistent with the documented automation and customization mechanisms (DWSIM Project, 2026a, 2026b).

The automation runner solves the native workspace, checks isotope activities against CPython, saves the file, reloads it and recalculates its operations. The native verification report, results/dwsim_verification.json, records the version, object count, solver status, numerical comparisons and SHA-256 of the saved file. The mathematical checks are recorded in results/verification.json. Reproduction commands, software versions and paths are provided in the accompanying README.

The research package contains original code, data transcriptions with source locations and derived results. It is publicly available at https://github.com/lucasfffaria/Ac225_DWSIM. Published article PDFs remain source literature and are not included in the repository. The reference list and provenance files provide their identifiers. All article findings are conditional on the model and input boundaries stated in the main text.

S7. Connected subflowsheet boundary, state variables and balances

The additional file dwsim/Ac225_Connected_Purification.dwxml contains seven custom operations, sixteen material streams and twenty-one native connections. Six operations form the connected primary-fraction-to-product sequence; the independent seventh operation evaluates the citrate/AG50 pH response. The file dwsim/Ac225_Isotope_Timing.dwxml remains a separate six-node baseline.

The streams carry five bulk analytical components: H_2O, HNO_3, HCl, NH_3 and NH_4Cl. They do not assert that these are the actual undissociated molecular species in solution. The total mass and the acid/base reaction are evaluated on the analytical-component basis. Nominal temperature and pressure are 298.15 K and 101325 Pa. DWSIM mass-flow values equal the batch component masses divided by 3600 s; this arbitrary reference throughput must not be interpreted as the actual 3 mL min^{-1} column flow. Reported column

pumping times use the latter separately. The mass–volume conversion uses the specified empirical density model.

Every stream stores the four isotope activities as A225_Bq, A226_Bq, A227_Bq and Ra225_Bq. It also stores age_d, batch_basis_s and batch_volume_mL. Radioactive trace masses are excluded from the macroscopic carrier balance. The primary boundary uses a nominal actinium recovery of 0.70 and starts with zero retained Ra. At the reference robust start of 12.64721 d, it advances the isotope clock by the fixed upstream five hours. Reagents have zero input activity. The supplied code can evaluate parent feeding in downstream operations, but the reported connected process cases use the parent-free boundary.

For a column, the component balance is

$$\mathbf{m}_{\mathrm{waste}} = \sum_{\mathrm{k}} \mathbf{m}_{\mathrm{inlet,k}} - \mathbf{m}_{\mathrm{cut}}. \tag{S19}$$

All components must remain nonnegative. The DGA2 cut is prescribed as 10 mL with scenario HCl carryover. The LN final cut is prescribed as 10 mL of 0.7 M HNO_3. These are collection specifications; the code does not claim that they follow from resin equilibrium or a hydrodynamic displacement calculation. Preconditioning liquid, loading effluent and washes are combined in batch-composite waste ledgers. The corresponding isotope waste represents rejection of the surviving trace inventory at the modeled operation endpoint, not a time-resolved elution chromatogram.

In the conditioner, NH_3 consumption is the limiting-reagent amount in $HCl + NH_3 \rightarrow NH_4Cl$. The model solves the base addition against final solution volume, including its change with composition. Default assumptions are 0.10 M HCl in the incoming DGA cut, 1 M aqueous NH_3, and 0.05 M residual analytical HCl. The resulting 0.47631 mL base addition contains 4.7631×10^{-4} mol NH_3. These values are scenarios because the source does not report the actual adjustment volume or pre-adjustment acid composition. No neutralization heat or pH is inferred.

Table S6. Connected reference-case stream inventories per batch. Volumes are correlation-based; no actual pumping-rate meaning attaches to the 3600 s reference basis. Product and composite waste values depend on the specified cuts.

Stream	Mass (g)	Volume (mL)	HNO_3 (mmol)	HCl (mmol)	^{225}Ac (kBq)
PRIMARY–FRACTION	103.651	80.000	800.000	0.000	285.093
DILUTED–FEED	183.414	158.035	800.000	0.000	284.955
DGA–CUT	9.988	10.000	0.000	1.000	270.704
LN–FEED	10.460	10.474	0.000	0.524	270.572
LN–CUT	10.210	10.000	7.000	0.000	257.721
FINAL–PRODUCT	10.210	10.000	7.000	0.000	256.028
DI–WATER	79.764	80.000	0.000	0.000	0.000
DGA–PRE	28.968	25.000	125.000	0.000	0.000
DGA–RINSE	16.428	15.000	0.000	90.000	0.000
DGA–ELUENT	9.970	10.000	0.000	0.000	0.000

NH3–ADDITION	0.472	0.476	0.000	0.000	0.000
LN–PRE	9.979	10.000	0.000	0.500	0.000
LN–RINSE	13.983	14.000	0.700	0.000	0.000
LN–ELUENT	10.210	10.000	7.000	0.000	0.000
DGA–WASTE	228.793	198.023	925.000	89.000	13.294
LN–WASTE	34.421	34.474	0.700	1.024	12.656

NH_3 and NH_4Cl inventories are included in the total masses and supplied in the full CSV. The primary-fraction value is an incoming boundary stream, not a fresh-reagent addition. Zero activity in a reagent indicates that no radioactivity was assigned to it in this scenario.

The complete reaction-adjusted component residual is less than 1.3×10^{-19} kg in CPython; the native global mass residual is 5.6×10^{-17} kg. Binary acid preparations, composite mixtures, neutralization and every native stream were checked on the same component order. The values establish numerical closure for the chosen boundary. They do not close the source experiment's omitted target-feed, sulfate, dead-volume, resin-retained or evaporation inventories.

The operation-level decay check uses the surviving Ac and Ra inventories plus a lumped heavy-descendant sink. Native operations compute scalar Bateman expressions; the external check propagates a matrix exponential. At equal chemical recovery for the three Ac isotopes, moving a parent-free transfer factor among stages changes product/waste allocation but not final product activity at a fixed age. The supplied factorization check varies DGA2 recovery from 0.92 to 1.00, adjusting LN recovery to preserve their product of 0.90857. The nominal factor of 0.95319 for each column is therefore explicitly non-identifiable from the source's final yield.

S8. Aqueous-property correlations and independent density check

Density is calculated from Laliberté's apparent-solute-density formulation (Laliberté, 2009):

$$\rho^{-1} = w_w/\rho_w + \sum_i w_i/\rho_{app,i} \tag{S20}$$

$$\rho_{app,i} = \frac{[c_{0,i}(1-w_w)+c_{1,i}]\exp\left[10^{-6}(t+c_{4,i})^2\right]}{(1-w_w)+c_{2,i}+c_{3,i}t} \tag{S21}$$

where t is temperature in degrees Celsius and the w values are mass fractions. The coefficients and validity limits for the four solutes are provided in data/liquid_property_parameters.json. They were transcribed from the open-source thermo 0.6.1 implementation, which attributes the data to Laliberté's 2009 compilation (Bell and contributors, 2026). Independently coded density and viscosity expressions agree with

that implementation to 6.2×10^{-16} relative across sixteen binary composition checks for each property. This is a software-verification check against the same published correlation, not an independent experimental test.

Viscosity uses the logarithmic mixing form $\ln \mu = w_w \ln \mu_w + \sum_i w_i \ln \mu_i$. The nitric-acid viscosity parameter set is limited to acid mass fraction 0.3086 and 4–25 °C. The hydraulic calculations remain at 25 °C and use the approximately 5 M diluted feed and the modeled column reagents. Viscosity of the 10 M primary fraction is not calculated with this correlation. The HCl, NH_3 and NH_4Cl solutions used in the screening also lie within their individual parameter ranges. Mixture-density and mixture-viscosity accuracy have not been established by the binary verification.

The external density check uses all 37 entries in the 298.15 K column of Table S1 of Hnedkovsky et al. (2025). The source reports density differences relative to water; these are compared directly with the predicted differences. Molality is converted to acid mass fraction as $w = mM/(1 + mM)$ using $M = 0.063013$ kg mol^{-1}. No coefficients are fitted to these data. The RMSE is 1.144 kg m^{-3}, the largest absolute density-difference residual is 3.229 kg m^{-3} and the largest relative solution-density deviation is 0.230%. The water reference used for the relative error is the correlation value of 997.0449 kg m^{-3}. The dataset was published after the 2009 correlation, providing an independent check of the binary nitric-acid density conversion over the required range.

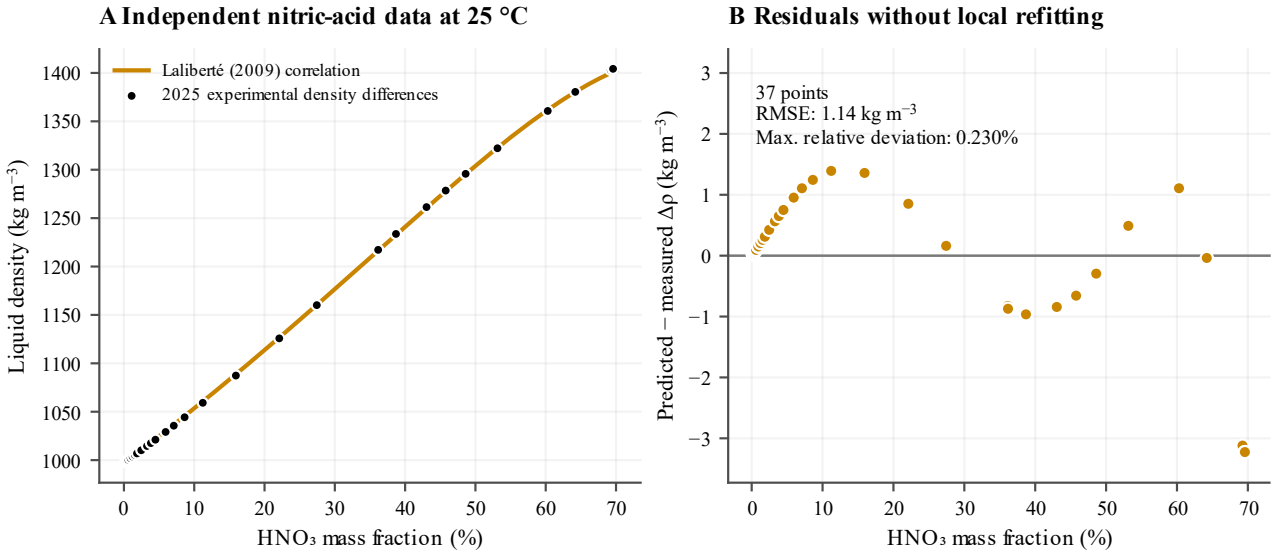


Figure S2. Independent binary nitric-acid density check at 298.15 K. Points use the density differences in Hnedkovsky et al., Table S1; the line is the earlier Laliberté correlation without refitting. This comparison does not test ternary mixtures, pH, enthalpy or chromatographic retention.

S9. Stage times, hydraulics and connection to the robust schedule

The DGA2 bed length is 52.61 mm and the LN bed length is 39.79 mm, calculated from reported bed volumes and diameters. Hydraulic input files distinguish these source values from the assumed effective

particle diameter and voidage. The pressure-envelope endpoints combine 150 µm with voidage 0.45 and 50 µm with voidage 0.35. The nominal case is 100 µm and 0.40. The largest pressure drop among the four sequential solutions is reported for each column. Frits, valves, tubing, resin compressibility, swelling, radiation damage and fouling are not included; no manufacturer pressure specification is inferred from the illustrative 50 kPa screen.

At the nominal flow, column pumping takes $208.03495/3 = 69.34498$ min for DGA2 and $44.47375/3 = 14.82458$ min for LN. The assigned dilution and adjustment operations add 20 min. A fixed allowance of 135.83043 min reconciles the sum to the reported 240 min downstream stage. This is a lumped timing convention. The 383.33 min difference for the entire partial nine-hour recipe includes a different boundary and remains unchanged. An additional queue is an explicit perturbation, not a reinterpretation of either unallocated duration as idle time.

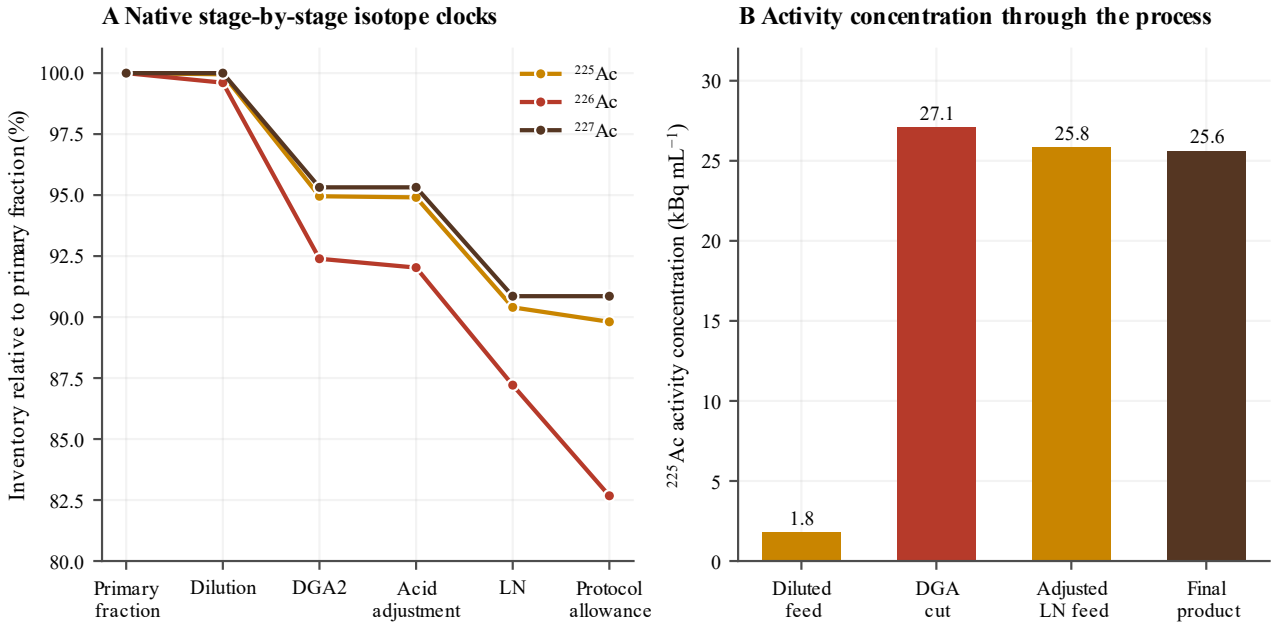


Figure S3. Isotope and activity-concentration histories in the connected reference case. A normalizes each isotope to its primary-fraction inventory; differing slopes between chemical cuts reflect half-life differences. B shows volumetric concentration after dilution and collection. Values use an arbitrary EOB-equivalent ^{225}Ac activity of 10^6 Bq, a start at 12.64721 d, nominal factorized chemical recovery and the declared operation-end transfer convention.

The native flow/queue sweep contains 25 recalculated cases, all checked against the external inventory propagation. The grid is provided in results/dwsim_process_sweep.csv. A finer CPython hydraulic scan supplements, rather than substitutes for, these native calculations. Doubling flow saves $(252.50870/3 - 252.50870/6) = 42.08478$ min. Its direct decay credit is $\exp[\lambda_{225}(42.08478/1440)] - 1 = 0.00204418$, conditional on an equally advanced permissible delivery. The relative recovery-loss budget is $1 - \exp[-\lambda_{225}(42.08478/1440)] = 0.00204001$.

The baseline external interval is obtained by subtracting 9/24 d from each endpoint of the original 3–5 d total interval. Adding an alternative deterministic processing duration to both endpoints preserves the

width. As long as the unconstrained optimum start is positive and no other scheduling constraints bind, the earliest robust start shifts to keep $c + D_- = L_+$. Therefore the worst delivery remains $L_+ + (D_+ - D_-)$, independent of that deterministic duration. Reducing the interval width by one day instead multiplies the guarantee by $\exp(\lambda_{225}) = 1.0723727$. This comparison keeps the same isotope box and recovery lower bound; it is not an estimate from observed delivery distributions or a multi-batch capacity model.

S10. Citrate/AG50 recovery response and interpolation sensitivity

The three fitted anchor values are 0.95 at pH 1.5, 0.98 at pH 2.0 and 0.66 at pH 2.5, reported for the 6 M HNO_3 fraction after loading in a 1 M citrate matrix (Fitzsimmons et al., 2019a). The source also reports approximately 98% loss in early fractions at pH 0.93; this is not treated as an exact product-recovery anchor and no extrapolation below pH 1.5 is made. The model is a local operating-response example for that published experiment. It is not a model of acid neutralization, and its pH variable must not be equated with the analytical 0.05 M acid target in the connected conditioner.

The linear model is $\eta(p) = 0.95 + 0.06(p - 1.5)$ for $1.5 \le p \le 2$, and $\eta(p) = 0.98 - 0.64(p - 2)$ for $2 < p \le 2.5$. For a symmetric pH bound δ whose optimized interval remains inside this domain, equalizing the two endpoint recoveries yields

$$p^* = 2 - \frac{0.64 - 0.06}{0.64 + 0.06}\delta \tag{S22}$$

$$\eta^*_{\min} = 0.98 - \frac{2(0.64)(0.06)}{0.64 + 0.06}\delta \tag{S23}$$

The code considers $0 \le \delta \le 0.25$. At $\delta = 0.20$, the linear setpoint is 1.83429 and the worst interpolated recovery is 0.958057, compared with 0.852 at setpoint 2.0. PCHIP gives a setpoint of 1.86972 and a minimum of 0.971353, compared with 0.91488 at 2.0. There is no experimental confidence interval for either interpolated optimum. This sensitivity is supplied to prevent the numerical setpoint from being mistaken for a validated control target.

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